GSM TECHNICAL SPECIFICATION

GSM 02.79

March 1999

Version 5.2.0

Source: SMG

Reference: RGTS/SMG-010279QR2

ICS: 33.020

Key words: Digital cellular telecommunications system, Global System for Mobile communications (GSM)

GLOBAL SYSTEM FOR
MOBILE COMMUNICATIONS

Digital cellular telecommunications system (Phase 2+); Support of Optimal Routeing (SOR); Service definition (Stage 1) (GSM 02.79 version 5.2.0 Release 1996)

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Foreword

This Global System for Mobile communications Technical Specification (GTS) was produced by the Special Mobile Group (SMG) Technical Committee of the European Telecommunications Standards Institute (ETSI).

This GTS specifies the stage 1 description of the first phase of Support of Optimal Routing (SOR) within the digital cellular telecommunications system.

GTS are produced by TC-SMG to enable the GSM Phase 2+ specifications to become publicly available, prior to submission for the formal ETSI standards approval procedure to become European Telecommunications Standards (ETS). This ensures the earliest possible access to GSM Phase 2+ specifications for all Manufacturers, Network operators and implementors of the Global System for Mobile communications.

The contents of this GTS are subject to continuing work within SMG and may change following formal SMG approval. Should TC-SMG modify the contents of this GTS it will then be republished by ETSI with an identifying change of release date and an increase in version number as follows:

Version 5.x.y
where:

- 5 indicates Release 1996 of GSM Phase 2+
- x the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- y the third digit is incremented when editorial only changes have been incorporated in the specification.

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1 Scope

This Stage 1 description of the first phase of Support of Optimal Routing (SOR):

- compiles the basic service requirements for SOR;
- describes the interactions for Supplementary Services (SS) in order to cater for SOR;
- refers to modifications to network features required by SOR.

This specification does not address the following:

- There is no need for optimisation of the routing of calls originally directed to a fixed network subscriber, because the physical address of a fixed network terminating line cannot differ from its logical address.
- SOR in non-GSM mobile networks is not a subject of this GTS, but might be possible by bilateral arrangement between GSM network operators and those non-GSM mobile network operators.

The purpose of SOR is to reduce the number of unnecessary inter-PLMN call legs.

The first phase of SOR applies to:

- OR for the benefit of the B party i.e. Mobile terminated calls with late call forwarding to the home or visited country (scenarios 1 and 2),

and optionally;

- OR for the benefit of the A party, e.g. Mobile to mobile calls where both mobile subscribers are in the same country (scenarios 3 to 10).

The complete set of scenarios included in the phase 1 of OR is presented in the paragraph for normal procedures. All other scenarios are excluded from Phase 1 of OR.

Note that Optimal Routing is applicable to national roaming situations, that is to calls directed to a mobile subscriber roaming in her home country, but registered in a PLMN different from her HPLMN.

All further call scenarios, including multiple call forwarding, are left to subsequent phases of SOR. Subsequent phases of SOR shall be backwards compatible with this first phase.

2 Normative references

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

This GTS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this GTS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] GSM 01.04 (ETR 350): "Digital cellular telecommunication system (Phase 2+); Abbreviations and acronyms".

3 Definitions and abbreviations

Administrative PLMN: The complete set of all functional entities normally regarded as a single PLMN.

Basic OR : Routeing of a call by the direct route when no forwarding occurs.

Direct route: A call uses the direct route if it is routed from the serving network of the original calling party to the serving network of the ultimate called party without any intermediate PLMN.

HPLMN route: A call uses the HPLMN route if the destination is deduced from the MSISDN of the called party. This forces the call to be routed via the HPLMN of the called party.

HPLMN leg: The HPLMN leg is that part of the HPLMN route from the IPLMN to the HPLMN of the called party.

IPLMN: An IPLMN is a PLMN which interrogates the HPLMN of a called party in order to determine the whereabouts of that party.

Early Call Forwarding : Early Call Forwarding is Call Forwarding performed from the IPLMN before the call has been extended to the VPLMN of the forwarding subscriber (i.e. Call Forwarding Unconditional and Conditional Call Forwarding on Not Reachable known at the IPLMN before extension of the call)

Functional PLMN: For the purposes of this description the Administrative PLMN is divided into functional PLMNs that represent different aspects of the Optimal Routeing functionality.

Late Call Forwarding: Late Call Forwarding is Call Forwarding performed after the call has been extended to the VPLMN of the forwarding subscriber (i.e. Conditional Call Forwarding on Busy, Conditional Call Forwarding on No Reply and Conditional Call Forwarding on Not Reachable detected in the VPLMN of the forwarding subscriber). Late Call Forwarding may be invoked in the IPLMN or in the VPLMN of the forwarding subscriber.

Special mobile network number: For the purposes of this description special mobile network numbers are numbers belonging logically to a PLMN but not to a mobile subscriber. Examples are the customer service number or value added service numbers.

Abbreviations are given in GSM 01.04 (ETR 350) [1]

4 Description of Optimal Routing

Support of Optimal Routing (SOR) is a network feature which enables the calls directed to a mobile subscriber to be routed directly to the mobile subscriber's actual location, or to her forwarded-to destination (instead of via the HPLMN or in the case of Late Call Forwarding via the VPLMN).

The IPLMN handling the call shall decide whether or not to optimise the routing of the call taking into account information provided by the called mobile subscriber's HPLMN. For given subscribers, as a network option, the HPLMN may permit or deny OR on a per call basis.

NOTE: References to the provision of Data Privacy supplementary services in case of OR have been deleted from GSM MoU reference documents.

5 Functional requirements

5.1 General

SOR shall be provided for all circuit switched Telecommunication Services, except emergency calls, Dedicated PAD and Dedicated Packet Access (GPRS is for further study).

The network feature shall be applied automatically for all calls except for those calls for which the HPLMN of the called party denies the optimal routing.

5.2 Normal operation

In the first phase of SOR, the routes of calls will be optimised for the cases of OR being performed within a country or towards the country where the call would have been routed normally.

Under all other circumstances, the GSM basic call routing is applied:

- If the IPLMN detects that basic OR cannot be applied, the IPLMN routes the call according to the dialled number.
- If the IPLMN detects that OR for Late Call Forwarding cannot be applied, the Call Forwarding will be performed in the VPLMN of the forwarding subscriber.

If the serving network of the ultimate destination supports SOR, the serving network of the caller shall be able to indicate to the network visited by the ultimate destination that the call has been optimally routed and to indicate the originating PLMN of the optimally routed leg.

The complete set of scenarios included in phase 1 of SOR is described below.

The notation used for the scenario diagrams is defined in figure 1.

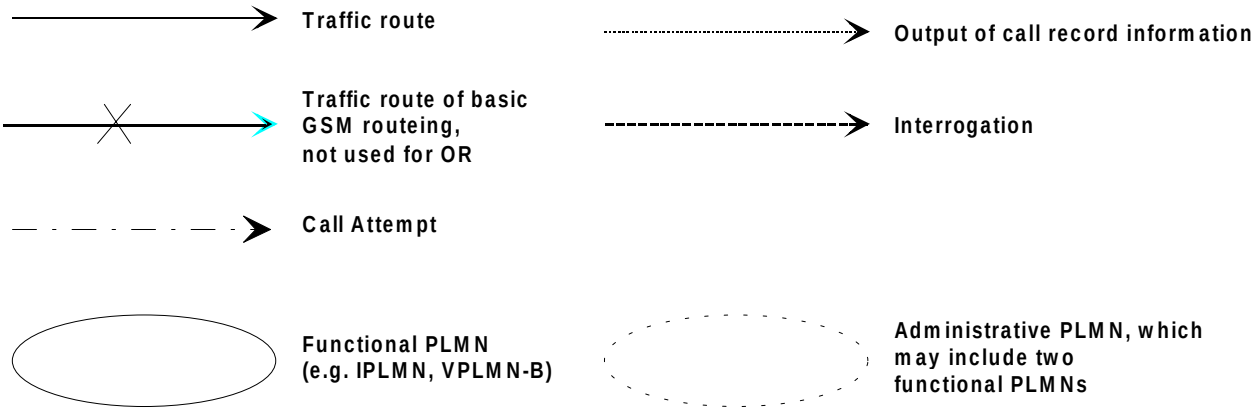


Figure 1: Notation used in scenario diagrams

If one of those scenarios occurs, SOR shall be invoked.

The assumptions taken are the following :

- A subscriber A sets up a call to a mobile subscriber B, who may have forwarded her calls to a subscriber C.
- The C subscriber may be either a fixed subscriber or a mobile subscriber.

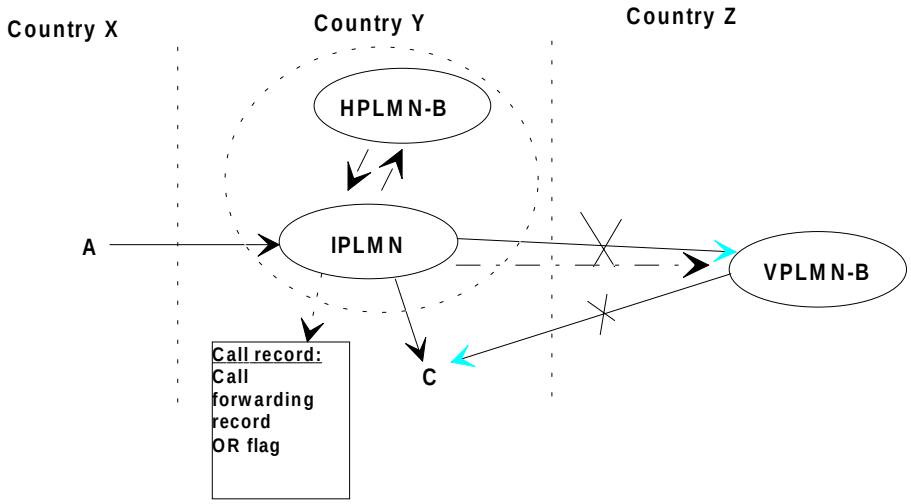
If C is a mobile subscriber, the location of C taken into account in the following paragraphs is the location of HPLMN-C.

5.2.1 A is a fixed subscriber or a mobile subscriber who may not benefit from OR

As the originating network does not have the ability to interrogate the HPLMN of the B subscriber, the normal GSM call handling is applied.

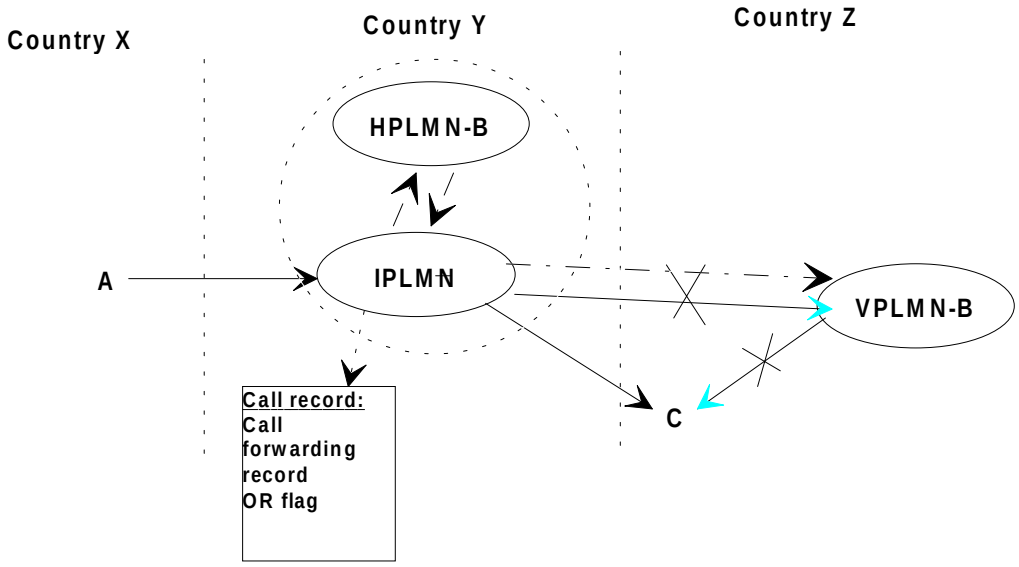
Once the HPLMN of the B subscriber has the control of the call, the call may be optimised in the case of a forwarded call towards a party located in the Home Country of the B subscriber or in the country visited by the B subscriber.

NOTE: - A may be in any country.
- In the two following cases, the IPLMN is HPLMN-B.



Scenario 1: OR for Late Call Forwarding, C is in the same country as HPLMN-B

NOTE: Call records shown in the figure are only for information



Scenario 2: OR for Late Call Forwarding, C is in the same country as VPLMN-B

NOTE: Call records shown in the figure are only for information

5.2.2 A is a mobile subscriber who may benefit from OR

5.2.2.1 Description of the call routing

If A sets up a call to B, then the originating PLMN shall interrogate the HPLMN of the mobile subscriber B in order to know how to route the call.

If B is registered in the same country as A, then the call shall be routed directly to B as described below.

If B has activated a Call Forward to a destination located in the Home Country of B or in the country where A is registered and this leads to the invocation of Early Call Forwarding, then the call shall be routed directly from A to the forwarded-to-party.

In any other cases, the call shall be routed to the mobile subscriber B via HPLMN(B).

If the call cannot be completed, the originating PLMN shall receive an indication of the reason of the failure of the call completion.

Thereafter, if the mobile subscriber B has activated a Conditional Call Forward to a destination located in her Home Country or to a destination in the country where A is registered and this leads to the invocation of Late Call Forwarding, then the originating PLMN shall route the call directly to the forwarded-to-party.

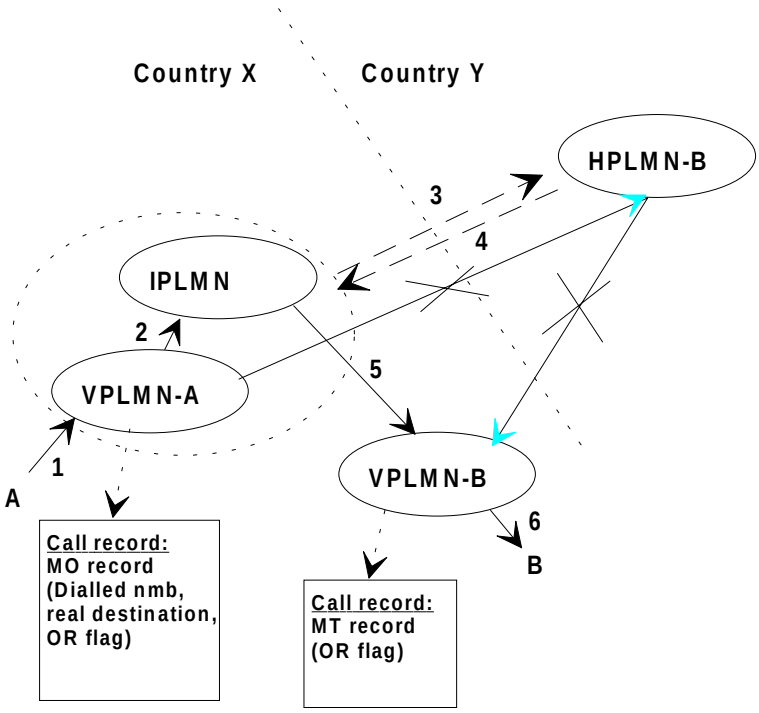
The remaining leg of the call from the intermediate point to the ultimate destination may be optimally routed.

5.2.2.2 Call scenarios handled in OR Phase 1

In the following scenarios, the IPLMN is VPLMN-A.

1) The call from the A subscriber to the B subscriber is completed (Basic OR)

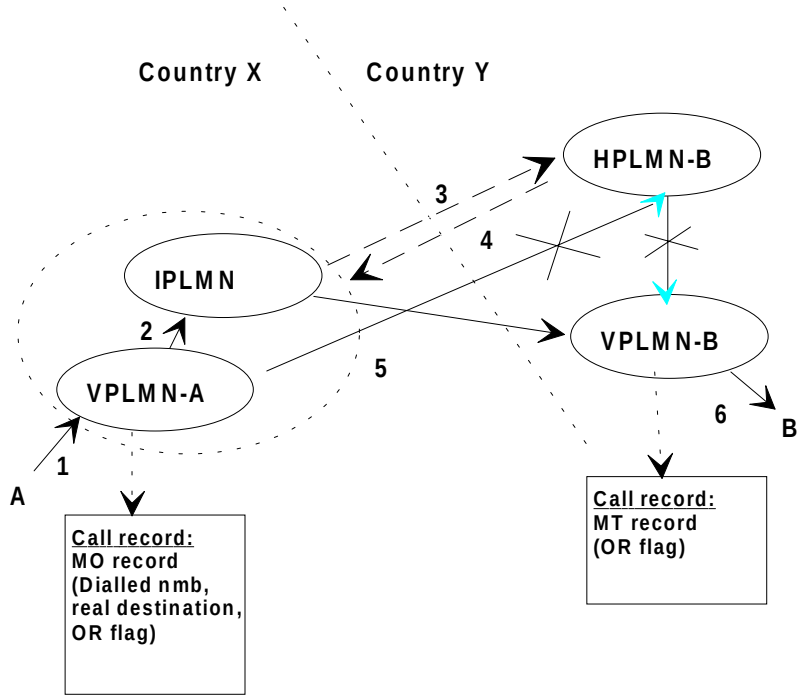
a) B is located in the same country as A



Scenario 3: BASIC OR, B in the same country as A

NOTE: Call records shown in the figure are only for information

b) B is located in her Home Country (Network usage optimization)



Scenario 4: BASIC OR, B in her home country

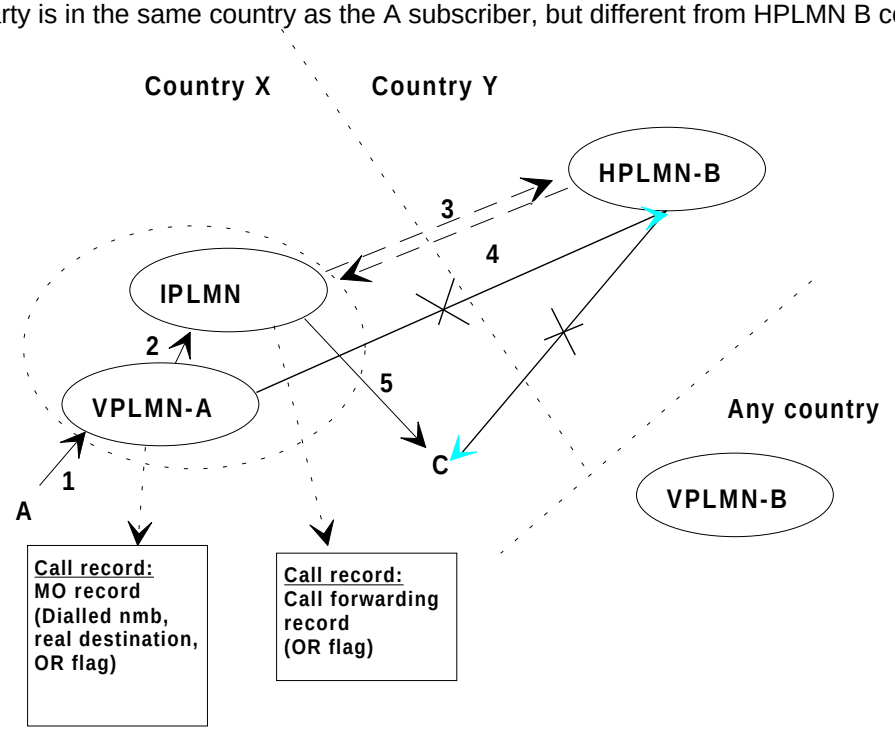
NOTE: Call records shown in the figure are only for information

2) The call set up by the subscriber A is being forwarded to a C party

2.1) Early Call Forward

NOTE: for the two following scenarios, the location of B is not relevant.

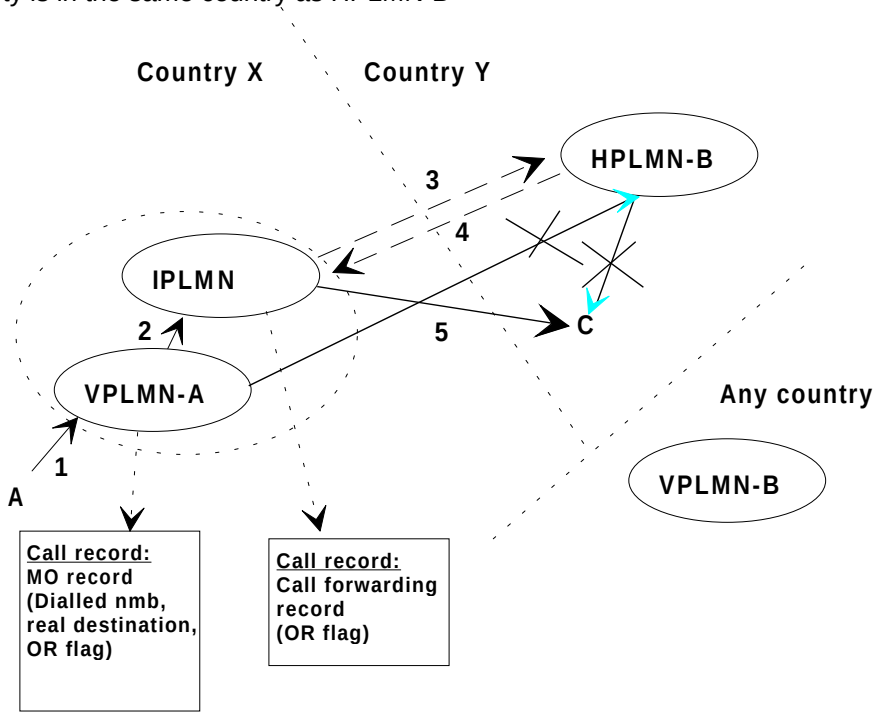
a) the C party is in the same country as the A subscriber, but different from HPLMN B country.



Scenario 5: Early CF, C in the same country as A

NOTE: Call records shown in the figure are only for information

b) the C party is in the same country as HPLMN-B

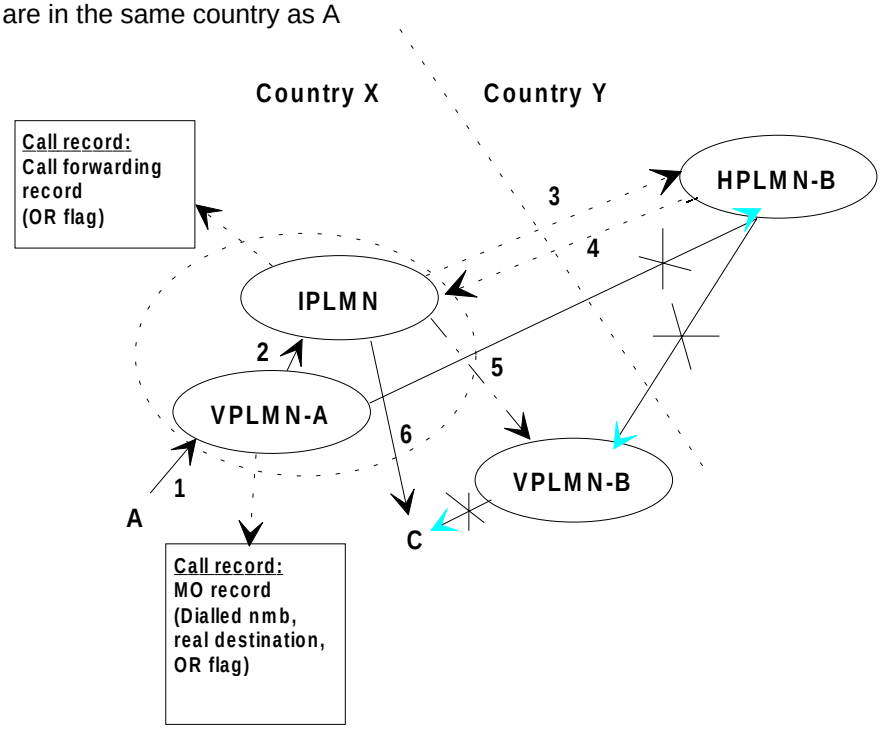


Scenario 6: Early CF, C in the same country as HPLMN-B

NOTE: Call records shown in the figure are only for information

2.2) Late Call Forward

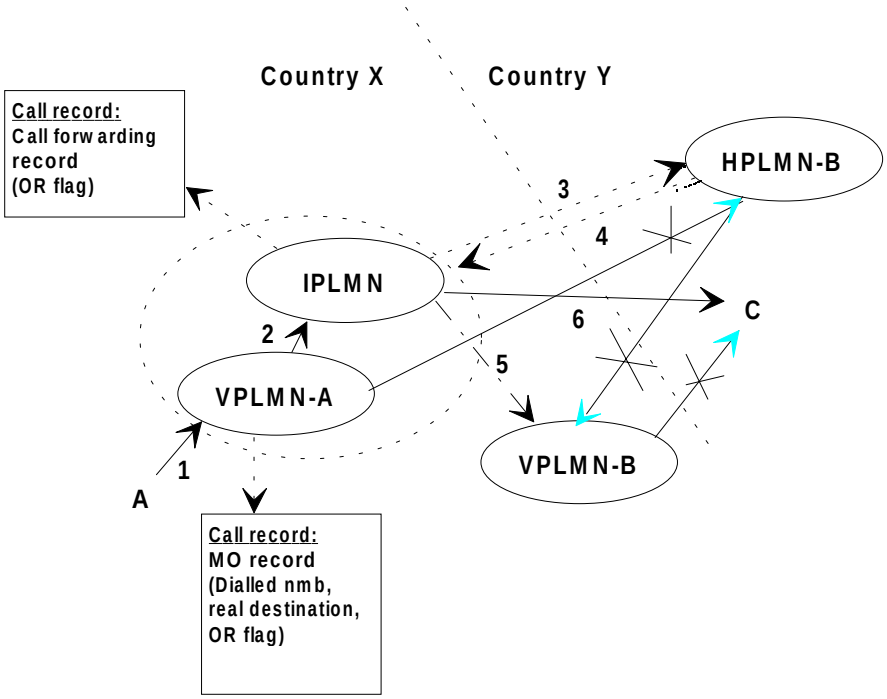
a) B and C are in the same country as A



Scenario 7: BASIC OR + OR for Late Call Forwarding, B in the same country as A, C in the same country as A

NOTE: Call records shown in the figure are only for information

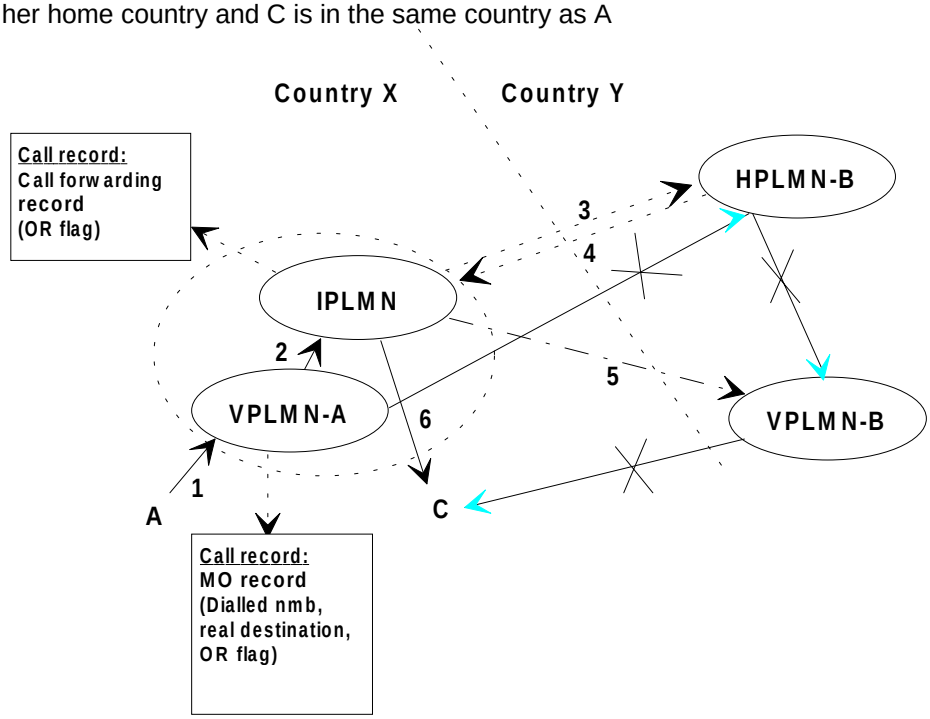
b) B is in the same country as A and C is in the same country as HPLMN-B



Scenario 8: BASIC OR + OR for Late Call Forwarding, B in the same country as A, C in the same country as HPLMN-B

NOTE: Call records shown in the figure are only for information

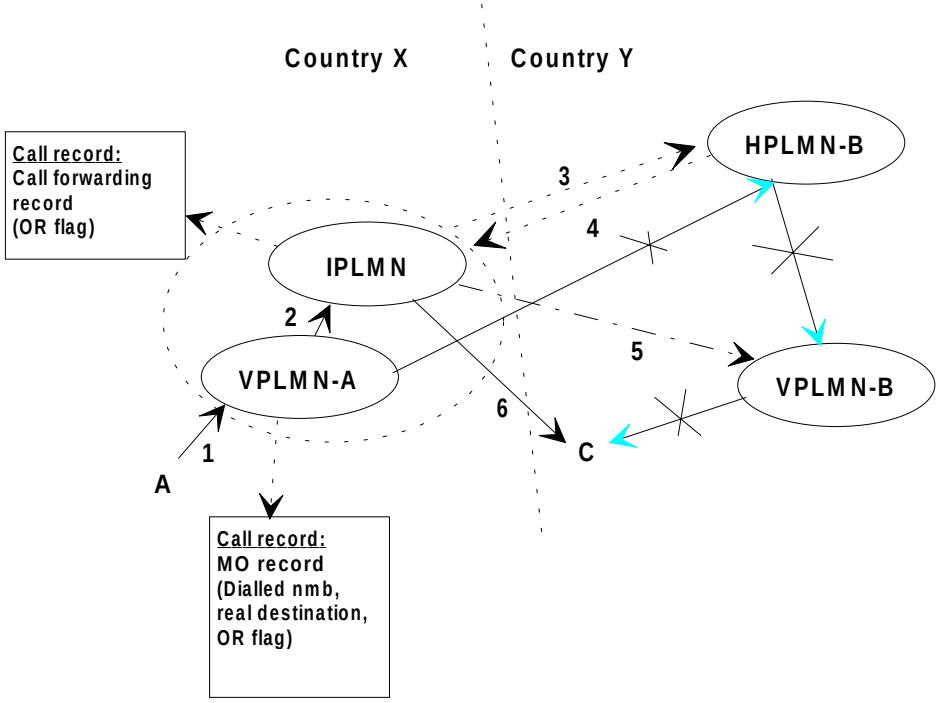
c) B is in her home country and C is in the same country as A



Scenario 9: BASIC OR + OR for Late Call Forwarding, B in the same country as HPLMN-B, C in the same country as A

NOTE: Call records shown in the figure are only for information

d) B and C are in the country of HPLMN-B



Scenario 10: BASIC OR + OR for Late Conditional Call Forwarding, B in the same country as HPLMN-B, C in the same country as HPLMN-B

NOTE: Call records shown in the figure are only for information

5.3 Exceptional procedures or unsuccessful outcome

5.3.1 Non-support of SOR in the VPLMN of the caller

If subscriber A roams into a network not supporting OR, she may not benefit from OR and only scenario 1 and scenario 2 of SOR are supported.

5.3.2 Non-support of SOR in the HPLMN of the B subscriber

If the HPLMN of the called party does not support SOR, SOR cannot be invoked and the route of the call as far as the called party's VPLMN cannot be optimised (the call would normally be routed via the HPLMN of the called party).

The same procedure shall be followed if the HPLMN operator denies OR on a subscriber basis.

5.3.3 Non support of SOR in the VPLMN of the B subscriber

If a subscriber whose HPLMN supports SOR registers in a VPLMN which does not support SOR, only the cases of Early Forwarded calls may be optimised.

5.3.4 Calls to special mobile network numbers

If the called number is a special mobile network number of the HPLMN, the HPLMN has to guarantee that the call terminates at its correct destination. This can be achieved by denying OR for this call by sending an appropriate error cause or by implementing a special handling.

6 Interactions of Optimal Routing with Supplementary Services

No Interaction unless it is stated differently below.

6.1 Call Forwarding

Interactions between SOR and Call Forwarding are dealt with in clause 5.0.

6.2 Call Barring

Outgoing Call Barring services are applied according to the dialled number.

Barring of Incoming calls when Roaming outside the HPLMN country will prevent calls to a mobile subscriber who has roamed outside her HPLMN country even if OR would result in no chargeable roaming leg.

The existing interactions between Call Forwardings and Call Barrings are not changed by the introduction of OR.

6.3 Call Transfer

A transferred call is considered as a set of two separate calls which may be separetely optimally routed.

Optimal Routing shall not be invoked as a result of the invocation of Call Transfer.

6.4 Call Deflection

A deflected call is considered as a late forwarded call and as such may be optimally routed.

6.5 Advice of Charge

Depending on call scenarios, AoC may not work properly.

7 Interactions of Optimal Routing with Operator Determined Barring (ODB)

The principles for the interaction between operator determined barring and SOR are the same as those for the interaction between Supplementary Service Call Barring and SOR.

8 Interactions of Optimal Routing with CAMEL

If CAMEL has to be applied to the Mobile Originating part of the call and if CAMEL modifies the destination of the call, the OR applies to any destination introduced by CAMEL.

If CAMEL has to be applied to the Mobile Terminating part of the call or a forwarding leg, the modified destination is treated for optimal routeing in the same way as a forwarded-to number.

If Barring services are also applicable to the call then they have to be handled first together with CAMEL before SOR is applied.

Annex A (informative): Change Request History

Change history					
SMG No.	**TDoc. No.**	**CR. No.**	**Section affected**	**New version**	**Subject/Comments**
SMG#19	366/96	None		2.0.0	Submitted for approval (Approved)
SMG#19	366/96	None		5.0.0	Approved at SMG#19
SMG#20	577/96	A001 A002 A003	8 3, 5.3.4 6.5	5.1.0	Clarifications on interactions of CAMEL and SOR Handling of special mobile network numbers when introducing SOR Interaction between Optimal Routeing and AoC
SMG#28	035/99	A004	1	5.2.0	Support of Optimal Routeing - Alignment to stage 2 & 3

History

Document history	
July 1996	Publication of Version 5.0.0
November 1996	Publication of Version 5.1.0
March 1999	Publication of Version 5.2.0

ISBN 2-7437-2929-5
Dépôt légal : Mars 1999